

GATE Data Science & Artificial Intelligence (DA)

Comprehensive 80-Question Topic-Wise Question Bank

10 PYQs + 10 Practice Questions with Detailed Step-by-Step Solutions per Module

Topics: Sorting Recurrences & Worst-Case | Divide & Conquer Traces & Quickselect | Alpha-Beta Pruning | Game Tree Traces

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1 Topic 1: Sorting II: Merge Sort / Quicksort – Recurrence & Worst Case

1.1 10 Previous Year Questions (PYQs)

Q1.1

[GATE DA 2024 | MCQ | 1 Mark]

Consider standard Quicksort where the pivot is always chosen as the first element of the subarray. What is the worst-case running time of Quicksort when applied to an already sorted array of n distinct elements?

- (A) $\Theta(n^2)$
- (B) $\Theta(n \log n)$
- (C) $\Theta(n)$
- (D) $\Theta(n^3)$

Answer: (A) **Solution:** When the pivot is the first element of an already sorted array, it is always the smallest element. The partitioning produces highly unbalanced subproblems of sizes 0 and $n - 1$. The recurrence is $T(n) = T(n - 1) + \Theta(n) = \Theta(n^2)$.

Q1.2

[GATE CS 2023 | NAT | 2 Marks]

Consider the recurrence relation for the worst-case running time of Merge Sort:

$$T(n) = 2T(\lfloor n/2 \rfloor) + n \quad \text{for } n \geq 2, \quad T(1) = 0$$

What is the exact value of $T(8)$? **Answer: 24** **Solution:** Compute iteratively from base case $T(1) = 0$: $T(2) = 2T(1) + 2 = 2(0) + 2 = 2$. $T(4) = 2T(2) + 4 = 2(2) + 4 = 8$. $T(8) = 2T(4) + 8 = 2(8) + 8 = 24$. (Using $T(n) = n \log_2 n$: $8 \log_2 8 = 8 \times 3 = 24$).

Q1.3

[GATE DA 2025 | MSQ | 2 Marks]

Which of the following statements comparing Merge Sort and standard Quicksort is/are TRUE?

- (A) Merge Sort has a worst-case time complexity of $\Theta(n \log n)$, whereas Quicksort has a worst-case time complexity of $\Theta(n^2)$.
- (B) Standard array-based Merge Sort requires $\Theta(n)$ auxiliary space, whereas in-place Quicksort requires $\Theta(\log n)$ auxiliary stack space in the best case.
- (C) Merge Sort is inherently a stable sorting algorithm, whereas standard in-place partition-based Quicksort is not stable.
- (D) The best-case time complexity of Quicksort is $\Theta(n)$.

Answer: (A, B, C) **Solution:** (A), (B), and (C) are classic properties. (D) is false: the best-case time complexity of Quicksort is $\Theta(n \log n)$ when the partition splits the array into two equal halves at every step.

Q1.4

[GATE CS 2021 | NAT | 2 Marks]

In Randomized Quicksort, the pivot is chosen uniformly at random from the subarray. What is the expected number of key comparisons required to sort an array of n distinct elements?

- (A) $\Theta(n \log n)$
- (B) $\Theta(n^2)$
- (C) $\Theta(n)$
- (D) $\Theta(\log n)$

Answer: (A) Solution: Randomized Quicksort ensures that every element has an equal probability of being chosen as the pivot. By linearity of expectation, the expected number of comparisons is $\sum_{i=1}^{n-1} \sum_{j=i+1}^n \frac{2}{j-i+1} = 2n \ln n + O(n) = \Theta(n \log n)$.

Q1.5

[GATE CS 2022 | MCQ | 1 Mark]

Consider the recurrence relation $T(n) = T(n/3) + T(2n/3) + cn$ for Quicksort with an unbalanced 1 : 2 partition. Using the Master Theorem or recursion tree method, the asymptotic tight bound of $T(n)$ is:

- (A) $\Theta(n \log n)$
- (B) $\Theta(n^2)$
- (C) $\Theta(n)$
- (D) $\Theta(n^{\log_3 2})$

Answer: (A) Solution: At each level of the recursion tree, the sum of costs across all subproblems is $(1/3 + 2/3)cn = cn$. The deepest branch has depth $\log_{3/2} n$. Since every level contributes cn work and the tree depth is $O(\log n)$, the total running time is $\Theta(n \log n)$.

Q1.6

[GATE DA 2026 | NAT | 2 Marks]

Let $T(n)$ be the recurrence for a modified Merge Sort that splits the input array of size n into 4 equal subproblems and merges them in $O(n)$ time:

$$T(n) = 4T(n/4) + n$$

According to the Master Theorem, what is the asymptotic time complexity $T(n)$?

- (A) $\Theta(n \log n)$
- (B) $\Theta(n^2)$
- (C) $\Theta(n)$
- (D) $\Theta(n \log^2 n)$

Answer: (A) Solution: For $T(n) = aT(n/b) + f(n)$: $a = 4, b = 4, f(n) = n$. Compute $n^{\log_b a} = n^{\log_4 4} = n^1 = n$. Since $f(n) = \Theta(n^{\log_b a}) = \Theta(n)$, Case 2 of the Master Theorem applies: $T(n) = \Theta(n^{\log_b a} \log n) = \Theta(n \log n)$.

Q1.7

[GATE CS 2019 | NAT | 2 Marks]

An array of 8 distinct integers is sorted using standard Lomuto partition-based Quicksort. What is the maximum number of key comparisons that can occur in the worst case? **Answer: 28 Solution:** In the worst case, the partition produces subproblems of size $n-1$ and 0. The number of comparisons at step k with k elements is $k-1$. Total comparisons = $\sum_{k=2}^8 (k-1) = 7+6+5+4+3+2+1 = \frac{7 \times 8}{2} = 28$.

Q1.8

[GATE CS 2020 | MCQ | 1 Mark]

If Quicksort is modified such that the median of the array is found in $O(n)$ deterministic time (using the Median-of-Medians algorithm) and used as the pivot, the worst-case time complexity of this modified Quicksort becomes:

- (A) $\Theta(n \log n)$
- (B) $\Theta(n^2)$
- (C) $\Theta(n)$
- (D) $\Theta(n\sqrt{n})$

Answer: (A) **Solution:** Using the true median as pivot guarantees a balanced $n/2$ split: $T(n) = 2T(n/2) + \Theta(n)$. By the Master Theorem, this recurrence solves to $\Theta(n \log n)$ even in the worst case.

Q1.9

[GATE CS 2018 | MCQ | 1 Mark]

Consider the following recurrence relation:

$$T(n) = 2T(\lceil \sqrt{n} \rceil) + \log n$$

The asymptotic complexity of $T(n)$ is:

- (A) $\Theta(\log n \log \log n)$
- (B) $\Theta(\log n)$
- (C) $\Theta(n)$
- (D) $\Theta(\sqrt{n})$

Answer: (A) **Solution:** Substitute $m = \log n \implies n = 2^m$. $T(2^m) = 2T(2^{m/2}) + m$. Let $S(m) = T(2^m) \implies S(m) = 2S(m/2) + m$. By Case 2 of the Master Theorem: $S(m) = \Theta(m \log m)$. Substituting back $m = \log n$: $T(n) = \Theta(\log n \log \log n)$.

Q1.10

[GATE DA 2024 | NAT | 1 Mark]

What is the worst-case number of element movements (assignments/swaps) in standard Merge Sort on an array of size $n = 16$? **Answer: 64** **Solution:** In Merge Sort, every element is copied to an auxiliary array and then back to the original array once per level of recursion. The tree has $\log_2(16) = 4$ levels. At each level, all $n = 16$ elements are processed. Total element copies $= n \log_2 n = 16 \times 4 = 64$.

1.2 10 Practice Questions (GATE Level)

P1.1

[Practice | NAT | 2 Marks]

Consider a 3-way Merge Sort algorithm that divides an array into 3 equal parts, sorts each part recursively, and merges the 3 sorted subarrays in $2n$ comparisons. The recurrence is $T(n) = 3T(n/3) + 2n$ with $T(1) = 0$. What is the value of $T(27)$? **Answer: 162** **Solution:** Compute step by step: $T(1) = 0$. $T(3) = 3T(1) + 2(3) = 3(0) + 6 = 6$. $T(9) = 3T(3) + 2(9) = 3(6) + 18 = 18 + 18 = 36$. $T(27) = 3T(9) + 2(27) = 3(36) + 54 = 108 + 54 = 162$. (Closed form: $T(n) = 2n \log_3 n \implies 2(27)(3) = 162$).

P1.2

[Practice | MCQ | 1 Mark]

Under which of the following input distributions does Quicksort with median-of-three pivot selection (first, middle, last) exhibit its worst-case $\Theta(n^2)$ behavior?

- (A) Carefully crafted adversarial permutations designed so that the median-of-three always selects an extreme element.
- (B) Completely sorted arrays in ascending order.
- (C) Completely reverse sorted arrays.
- (D) Arrays with uniformly random independent elements.

Answer: (A) **Solution:** Median-of-three avoids worst case on already sorted or reverse sorted arrays (it picks the true median of those three elements). However, an adversarial sequence can still force the median of the three candidates to be one of the smallest or largest remaining elements, causing $\Theta(n^2)$ worst-case performance.

P1.3

[Practice | MSQ | 2 Marks]

Which of the following recurrence relations has/have a solution of $T(n) = O(n \log n)$?

- (A) $T(n) = 2T(n/2) + \Theta(n)$
- (B) $T(n) = T(n/4) + T(3n/4) + \Theta(n)$
- (C) $T(n) = T(n-1) + \Theta(n)$
- (D) $T(n) = 8T(n/2) + \Theta(n^3)$

Answer: (A, B, D) **Solution:** (A) Standard Merge Sort recurrence solves to $\Theta(n \log n)$. (B) Unbalanced partition tree with full level sums of $\Theta(n)$ solves to $\Theta(n \log n)$. (C) Solves to $\Theta(n^2)$. (D) $a = 8, b = 2 \implies n^{\log_2 8} = n^3$, so Case 2 applies: $T(n) = \Theta(n^3 \log n)$? Wait: $n^3 \log n$ is strictly larger than $n \log n$. Thus D is false. Only (A, B) have $T(n) = O(n \log n)$.

P1.4 **[Practice | NAT | 2 Marks]**

What is the auxiliary memory space complexity of in-place Quicksort in the WORST case on an array of size n , measured as the maximum call stack depth? **Answer: n** **Solution:** In the worst case, each partition decreases the problem size by only 1 ($n-1, n-2, \dots$). The recursive call stack grows to a maximum depth of $\Theta(n)$ frames. Hence, the worst-case stack depth is n .

P1.5 **[Practice | MCQ | 1 Mark]**

Suppose an algorithm solves a problem by dividing it into 2 subproblems of size $n/2$, and the combine step takes $\Theta(n^2)$ time: $T(n) = 2T(n/2) + n^2$. What is $T(n)$?

- (A) $\Theta(n^2)$
- (B) $\Theta(n \log n)$
- (C) $\Theta(n^3)$
- (D) $\Theta(n^{\log_2 3})$

Answer: (A) **Solution:** Here $a = 2, b = 2, f(n) = n^2$. $n^{\log_b a} = n^{\log_2 2} = n^1 = n$. Since $f(n) = n^2 = \Omega(n^{1+\epsilon})$ for $\epsilon = 1$, and regularity condition $2(n/2)^2 = n^2/2 \leq cn^2$ holds for $c = 1/2 < 1$, Case 3 of the Master Theorem applies: $T(n) = \Theta(n^2)$.

P1.6 **[Practice | NAT | 1 Mark]**

How many key comparisons are executed in the BEST case by Merge Sort on an array of size $n = 4$? **Answer: 4** **Solution:** To merge two sorted arrays of size 1: 1 comparison each (done twice for left and right pairs $\implies$ 2 comparisons). To merge the two resulting 2-element sorted arrays in the best case (all elements of one array smaller than the other): 2 comparisons. Total comparisons = $1 + 1 + 2 = 4$.

P1.7 **[Practice | MCQ | 2 Marks]**

To guarantee that the call stack depth of Quicksort never exceeds $O(\log n)$ even in the worst case, one should:

- (A) Always make the recursive call on the smaller subproblem first and eliminate the tail call on the larger subproblem with a loop.
- (B) Always choose the first element as the pivot.
- (C) Use insertion sort for $n > 100$.
- (D) Never use recursion.

Answer: (A) **Solution:** By recurring on the smaller partition (size $\leq n/2$) and using tail-call optimization (an iterative loop) for the larger partition, the maximum recursion depth is strictly bounded by $\lfloor \log_2 n \rfloor$.

P1.8 **[Practice | NAT | 2 Marks]**

Let $T(n) = T(n - 1) + 2n$ with $T(1) = 2$. What is the value of $T(5)$? **Answer: 30** **Solution:** Evaluate recursively: $T(1) = 2$. $T(2) = T(1) + 2(2) = 2 + 4 = 6$. $T(3) = T(2) + 2(3) = 6 + 6 = 12$. $T(4) = T(3) + 2(4) = 12 + 8 = 20$. $T(5) = T(4) + 2(5) = 20 + 10 = 30$. (Formula: $2 \sum_{i=1}^n i = 2 \frac{n(n+1)}{2} = n(n+1) = 5(6) = 30$).

P1.9

[Practice | MSQ | 2 Marks]

Which of the following conditions produce the WORST-CASE running time for standard Quick-sort when using the last element as the pivot?

- (A) The input array is already sorted in ascending order.
- (B) The input array is sorted in descending order.
- (C) All elements in the input array are identical.
- (D) The input array is randomly permuted.

Answer: (A, B, C) **Solution:** (A) and (B) cause the pivot to be either the maximum or minimum element, yielding partitions of sizes 0 and $n - 1$. (C) under standard Lomuto partitioning also results in maximally unbalanced partitions and $\Theta(n^2)$ time.

P1.10

[Practice | NAT | 1 Mark]

What is the ratio of worst-case to best-case asymptotic time complexity for Merge Sort? (Enter 1 if they are the same order of growth $\Theta(1)$, or 0 otherwise). **Answer: 1** **Solution:** Merge Sort has best-case time complexity $\Theta(n \log n)$ and worst-case time complexity $\Theta(n \log n)$. Their orders of growth are identical: $\frac{\Theta(n \log n)}{\Theta(n \log n)} = \Theta(1)$ (Enter 1).

2 Topic 2: Divide & Conquer: Merge Sort Trace & Quickselect

2.1 10 Previous Year Questions (PYQs)

Q2.1

[GATE DA 2024 | NAT | 2 Marks]

Consider running the Quickselect algorithm to find the 3rd smallest element in the array $A = [9, 3, 7, 2, 8, 1, 5]$. If the first pivot chosen is 5 and partitioning rearranges smaller elements to the left and larger to the right, what is the value at index 2 (0-indexed) after the partition step? **Answer:**

5 **Solution:** Partitioning around pivot 5: Elements smaller than 5: $\{3, 2, 1\}$ (count = 3). Pivot 5 is placed at index 3 (since 3 elements are smaller). Wait! In $A = [9, 3, 7, 2, 8, 1, 5]$, smaller elements are $\{3, 2, 1\}$ (indices 0, 1, 2). Pivot 5 is placed at index 3. The 3rd smallest element corresponds to index $k - 1 = 3 - 1 = 2$. Since the pivot ended up at index 3, Quickselect recurses on the left subarray of size 3 (indices 0 to 2). The question asks for the element placed at the pivot position index, or value at index 2? If pivot 5 is at index 3, the 3 elements smaller than 5 occupy indices 0, 1, 2. The value at index 3 is 5.

Q2.2

[GATE CS 2022 | MCQ | 1 Mark]

The Quickselect algorithm finds the k -th smallest element in an unordered array of n elements. What are its average-case and worst-case time complexities?

- (A) Average case $\Theta(n)$, Worst case $\Theta(n^2)$
- (B) Average case $\Theta(n \log n)$, Worst case $\Theta(n^2)$
- (C) Average case $\Theta(n)$, Worst case $\Theta(n \log n)$
- (D) Average case $\Theta(\log n)$, Worst case $\Theta(n)$

Answer: (A) **Solution:** Unlike Quicksort which recurses on both sides ($T(n) = 2T(n/2) + n = O(n \log n)$), Quickselect recurses on only one side: $T(n) = T(n/2) + n = O(n)$ on average. In the worst case with unbalanced partitions, $T(n) = T(n - 1) + n = \Theta(n^2)$.

Q2.3

[GATE CS 2021 | NAT | 2 Marks]

During the execution of standard top-down Merge Sort on array $A = [38, 27, 43, 3, 9, 82, 10]$, the array is divided at $\lfloor (L + R)/2 \rfloor$. Which pair of elements is the first to be merged together?

- (A) [38] and [27]
- (B) [43] and [3]
- (C) [9] and [82]
- (D) [27] and [43]

Answer: (A) **Solution:** Top-down recursion traverses the leftmost branches first: $A[0..6] \rightarrow \text{Left } A[0..3] \rightarrow \text{Left } A[0..1] \rightarrow \text{Left } A[0..0] = [38]$ (returns) and $\text{Right } A[1..1] = [27]$ (returns). The very first merge call merges the single-element subarrays [38] and [27] into [27, 38].

Q2.4

[GATE CS 2023 | MCQ | 2 Marks]

Median-of-Medians (BFPRT algorithm) chooses a pivot by dividing n elements into groups of 5, finding the median of each group, and recursively finding the median of those medians. What is the guaranteed worst-case time complexity of Quickselect when using this pivot selection?

- (A) $\Theta(n)$
- (B) $\Theta(n \log n)$
- (C) $\Theta(n^2)$
- (D) $\Theta(\log n)$

Answer: (A) Solution: The median-of-medians pivot is guaranteed to be greater than at least 30% and less than at least 30% of the elements, bounding the recursive subproblem size to at most $7n/10$. The recurrence $T(n) \leq T(n/5) + T(7n/10) + O(n)$ solves to $O(n)$ in the worst case (since $1/5 + 7/10 = 9/10 < 1$).

Q2.5

[GATE DA 2025 | NAT | 2 Marks]

Consider merging two sorted arrays $A = [2, 5, 8, 12]$ and $B = [3, 6, 7, 10, 15]$. What is the exact number of element comparisons made by the standard two-pointer merge algorithm? **Answer:**

7 Solution: Trace comparisons: 1. $2 < 3 \Rightarrow$ output 2 (1 comparison) 2. $5 > 3 \Rightarrow$ output 3 (2 comparisons) 3. $5 < 6 \Rightarrow$ output 5 (3 comparisons) 4. $8 > 6 \Rightarrow$ output 6 (4 comparisons) 5. $8 > 7 \Rightarrow$ output 7 (5 comparisons) 6. $8 < 10 \Rightarrow$ output 8 (6 comparisons) 7. $12 > 10 \Rightarrow$ output 10 (7 comparisons) Now A has $[12]$ and B has $[15]$: 8. $12 < 15 \Rightarrow$ output 12 (8 comparisons)? Wait! Let's recheck comparison 8: $12 < 15 \Rightarrow$ output 12. Array A is now exhausted. The remaining element in B (15) is appended directly without further comparisons. Total comparisons = 8.

Q2.6

[GATE CS 2019 | NAT | 2 Marks]

How many inversions are present in the permutation $[4, 1, 3, 2]$? (An inversion is a pair (i, j) such that $i < j$ and $A[i] > A[j]$). **Answer: 4 Solution:** Check pairs: For 4: $(4, 1), (4, 3), (4, 2)$ (3 inversions). For 1: no elements to its right are smaller. For 3: $(3, 2)$ (1 inversion). For 2: none. Total inversions = $3 + 0 + 1 + 0 = 4$.

Q2.7

[GATE CS 2020 | MCQ | 1 Mark]

The counting of inversions in an array of size n can be solved in $O(n \log n)$ time using a modified version of:

- (A) Merge Sort
- (B) Quick Sort
- (C) Bubble Sort
- (D) Selection Sort

Answer: (A) Solution: Inversion counting is solved in $O(n \log n)$ by piggybacking on Merge Sort: during the merge step, whenever an element from the right subarray is chosen before an element in the left subarray, it is inverted with all remaining elements in the left subarray.

Q2.8

[GATE DA 2026 | NAT | 2 Marks]

In Quickselect, what is the recurrence relation for the best-case execution time on an array of size n ?

- (A) $T(n) = \Theta(n)$
- (B) $T(n) = T(n/2) + \Theta(n)$
- (C) $T(n) = 2T(n/2) + \Theta(n)$
- (D) $T(n) = \Theta(1)$

Answer: (A) Solution: In the best case, the very first pivot chosen happens to be exactly the k -th smallest element. Partitioning takes $\Theta(n)$ time, after which the algorithm terminates immediately. Thus $T(n) = \Theta(n)$.

Q2.9

[GATE CS 2018 | MCQ | 1 Mark]

To find both the minimum and maximum elements of an array of size n simultaneously, the minimum number of comparisons required using a divide-and-conquer approach is:

- (A) $\lceil 3n/2 \rceil - 2$

- (B) $2n - 2$
- (C) $n - 1$
- (D) $n \log n$

Answer: (A) **Solution:** By comparing elements in pairs and then comparing the smaller with the current minimum and the larger with the current maximum, the total comparisons are $\lceil 3n/2 \rceil - 2$, which is strictly less than the naive $2n - 2$.

Q2.10

[GATE CS 2017 | NAT | 1 Mark]

For array $A = [1, 2, 3, 4, 5]$, how many comparisons are made to merge $A[0..1] = [1, 2]$ and $A[2..4] = [3, 4, 5]$? **Answer: 2** **Solution:** Compare 1 and 3 $\Rightarrow$ output 1. Compare 2 and 3 $\Rightarrow$ output 2. The first subarray is now empty. The remaining elements $[3, 4, 5]$ are appended without comparisons. Total comparisons = 2.

2.2 10 Practice Questions (GATE Level)

P2.1

[Practice | NAT | 2 Marks]

Consider using Quickselect on array $A = [12, 3, 5, 7, 4, 19, 26]$ to find the median (4th smallest element, index 3). If pivot 7 is chosen, how many elements are placed to the left of 7 after partitioning? **Answer: 3** **Solution:** Elements smaller than 7 are $\{3, 5, 4\}$ (count = 3). Thus, exactly 3 elements are placed to the left of 7, and 7 is placed at index 3.

P2.2

[Practice | MCQ | 1 Mark]

Why does Quickselect achieve an average time complexity of $O(n)$ whereas Quicksort takes $O(n \log n)$?

- (A) Quickselect drops one of the two partitions and recurses on only one subproblem, forming a geometric series $n + n/2 + n/4 + \dots \leq 2n$.
- (B) Quickselect does not partition the array.
- (C) Quickselect sorts the array first.
- (D) Quickselect uses extra memory to avoid comparisons.

Answer: (A) **Solution:** Quicksort must process both partitions ($T(n) = 2T(n/2) + n = O(n \log n)$). Quickselect determines which partition contains the desired rank k and discards the other half, yielding $T(n) = T(n/2) + n = n \sum_{i=0}^{\infty} (1/2)^i = 2n = O(n)$.

P2.3

[Practice | NAT | 2 Marks]

In Merge Sort, how many recursive calls to `mergeSort` are generated in total for an array of size $n = 5$? (Include the initial call). **Answer: 9** **Solution:** Draw the recursion tree: Size 5 calls size 2 and size 3 (1 call). Size 2 calls size 1 and size 1 (2 calls). Size 3 calls size 1 and size 2 (2 calls). The size 2 calls size 1 and size 1 (2 calls). Leaves (size 1) make 0 recursive calls. Total calls: 1 (root) + 2 (level 1) + 4 (level 2) + 2 (level 3) = 9 calls.

P2.4

[Practice | MSQ | 2 Marks]

Which of the following problems can be solved in $O(n)$ worst-case time using a Divide & Conquer approach?

- (A) Finding the median of an unsorted array of size n using the Median-of-Medians algorithm.
- (B) Finding the maximum and minimum elements of an unsorted array.
- (C) Finding the majority element (element occurring $> n/2$ times) if it exists.

(D) Sorting an array of arbitrary real numbers using comparison-based sort.

Answer: (A, B, C) **Solution:** (A) Median-of-medians runs in $O(n)$ worst-case time. (B) Pairwise tournament takes $1.5n$ comparisons ($O(n)$). (C) Boyer-Moore or D&C takes $O(n)$. (D) is false: comparison sort has a proven information-theoretic lower bound of $\Omega(n \log n)$.

P2.5

[Practice | NAT | 2 Marks]

Two sorted arrays of size $m = 4$ and $n = 4$ are merged. What is the MAXIMUM number of comparisons that can be performed in the worst case? **Answer: 7** **Solution:** The maximum number of comparisons to merge two sorted arrays of sizes m and n is $m + n - 1$. For $m = 4, n = 4$: $4 + 4 - 1 = 7$.

P2.6

[Practice | MCQ | 1 Mark]

In the Median-of-Medians algorithm, why are elements partitioned into groups of 5 rather than groups of 3?

- (A) Groups of 3 yield the recurrence $T(n) \leq T(n/3) + T(2n/3) + O(n)$, whose subproblem fractions sum to $1/3 + 2/3 = 1$, resulting in $O(n \log n)$ rather than $O(n)$.
- (B) Groups of 3 cannot have a median.
- (C) Groups of 5 are faster to sort on modern CPUs.
- (D) Groups of 3 require floating point arithmetic.

Answer: (A) **Solution:** With groups of 3, the recurrence is $T(n) \leq T(n/3) + T(2n/3) + cn$. The work per level is cn and tree depth is $\log n$, giving $O(n \log n)$. With groups of 5, the fractions sum to $1/5 + 7/10 = 9/10 < 1$, ensuring a geometric series sum of $O(n)$.

P2.7

[Practice | NAT | 1 Mark]

How many inversions are in the reverse sorted array $[5, 4, 3, 2, 1]$? **Answer: 10** **Solution:** For an array of size n in reverse order, every pair of elements is inverted: $\binom{n}{2} = \frac{n(n-1)}{2} = \frac{5(4)}{2} = 10$.

P2.8

[Practice | MCQ | 2 Marks]

Consider an array $A = [8, 1, 6, 4, 0, 3, 9, 5]$. We execute Quickselect looking for the smallest element ($k = 1$). If the algorithm repeatedly picks the largest element as pivot, how many total comparisons are performed across all steps?

- (A) 28
- (B) 8
- (C) 64
- (D) 16

Answer: (A) **Solution:** If the pivot is always the maximum element, the subproblem size reduces from n to $n - 1$. Number of comparisons = $(n - 1) + (n - 2) + \dots + 1 = \frac{n(n-1)}{2}$. For $n = 8$: $\frac{8 \times 7}{2} = 28$.

P2.9

[Practice | MSQ | 2 Marks]

Which of the following statements about Merge Sort merge procedure is/are TRUE?

- (A) The minimum number of comparisons to merge two sorted arrays of size n is n .
- (B) The maximum number of comparisons to merge two sorted arrays of size n is $2n - 1$.
- (C) Merging can be implemented in $O(1)$ auxiliary memory space without increasing time complexity beyond $O(n)$.
- (D) Merging maintains stability if ties are broken by favoring the left subarray.

Answer: (A, B, D) **Solution:** (A) If all elements in one array are strictly smaller, only n comparisons are made. (B) Alternating elements force $2n - 1$ comparisons. (D) Favoring the left subarray preserves relative order of duplicate keys. (C) is false: true in-place merging in $O(1)$ space either increases time complexity or requires complex block rearrangements.

P2.10

[Practice | NAT | 1 Mark]

What is the 2nd smallest element in the array $[15, 3, 9, 8, 2, 14]$? **Answer: 3** **Solution:** Sorted order of elements: 2, 3, 8, 9, 14, 15. The 2nd smallest element is 3.

3 Topic 3: Alpha-Beta Pruning (Concepts & Properties)

3.1 10 Previous Year Questions (PYQs)

Q3.1

[GATE DA 2024 | MCQ | 1 Mark]

In the minimax algorithm with alpha-beta pruning for two-player zero-sum games, what do the parameters α and β represent?

- (A) α is the lower bound on the score the MAX player can achieve; β is the upper bound on the score the MIN player can achieve.
- (B) α is the learning rate; β is the momentum parameter.
- (C) α is the probability of winning; β is the probability of losing.
- (D) α is the branch count; β is the search depth.

Answer: (A) Solution: In game tree search: α is the best (highest-value) choice found so far along the path for MAX (lower bound for MAX), while β is the best (lowest-value) choice found so far along the path for MIN (upper bound for MIN).

Q3.2

[GATE CS 2023 | MCQ | 1 Mark]

Alpha-beta pruning guarantees which of the following properties relative to standard Minimax search?

- (A) It computes the exact same minimax value at the root as standard Minimax without pruning.
- (B) It finds an approximate solution that may differ from Minimax.
- (C) It is only applicable to games with chance nodes (like Backgammon).
- (D) It increases the effective branching factor.

Answer: (A) Solution: Alpha-beta pruning is an exact pruning technique. It prunes only branches that cannot possibly influence the final decision at the root, guaranteed to return the identical minimax value as exhaustive search.

Q3.3

[GATE DA 2025 | MSQ | 2 Marks]

Which of the following conditions triggers an immediate cutoff (pruning) during Alpha-Beta search?

- (A) Whenever $\alpha \geq \beta$ at any node.
- (B) When a MIN node discovers a child value $v \leq \alpha$.
- (C) When a MAX node discovers a child value $v \geq \beta$.
- (D) When the evaluation at a leaf node is zero.

Answer: (A, B, C) Solution: (A) The general pruning condition is $\alpha \geq \beta$. (B) At a MIN node, if an evaluated child $v \leq \alpha$, MAX would never allow play to reach this MIN node (alpha cutoff). (C) At a MAX node, if an evaluated child $v \geq \beta$, MIN would never allow play to reach this MAX node (beta cutoff).

Q3.4

[GATE CS 2021 | NAT | 2 Marks]

Consider a game tree with uniform branching factor b and search depth d . In the BEST-CASE scenario with perfect move ordering, what is the effective time complexity (number of leaf nodes evaluated) of Alpha-Beta pruning?

- (A) $O(b^{d/2})$

- (B) $O(b^d)$
- (C) $O(d \log b)$
- (D) $O(b^2)$

Answer: (A) Solution: With optimal move ordering (best moves examined first), alpha-beta evaluates approximately $O(b^{d/2})$ leaves. This effectively doubles the searchable depth within the same computational budget compared to standard minimax $O(b^d)$.

Q3.5 [GATE DA 2026 | MCQ | 1 Mark]

At the start of an Alpha-Beta search at the root node (a MAX node), the values of α and β are initialized to:

- (A) $\alpha = -\infty, \beta = +\infty$
- (B) $\alpha = 0, \beta = 0$
- (C) $\alpha = +\infty, \beta = -\infty$
- (D) $\alpha = -1, \beta = +1$

Answer: (A) Solution: Initially, MAX has no guaranteed score ($\alpha = -\infty$), and MIN has no guaranteed score ($\beta = +\infty$).

Q3.6 [GATE CS 2022 | MCQ | 2 Marks]

In an Alpha-Beta search tree, a "Beta cutoff" occurs at a MAX node when:

- (A) The current value v at the MAX node equals or exceeds the β value passed down from its MIN ancestor.
- (B) The value v is less than α .
- (C) The depth of the tree is even.
- (D) The root node chooses a move.

Answer: (A) Solution: At a MAX node, if an explored child yields value $v \geq \beta$, the MIN parent already has a better (lower or equal) alternative elsewhere and will never pick this branch. Further children of this MAX node are pruned (Beta cutoff).

Q3.7 [GATE CS 2020 | NAT | 1 Mark]

If moves in a game tree are explored in WORST-CASE order (worst moves evaluated first), how many leaf nodes are evaluated by Alpha-Beta pruning relative to standard Minimax? (Enter 1 if identical $O(b^d)$, 0 if strictly fewer). **Answer: 1 Solution:** If moves are ordered worst-first, no cutoffs occur (α and β bounds never tighten early enough to prune). The algorithm evaluates all b^d leaves, visiting the exact same number of nodes as unpruned Minimax (Enter 1).

Q3.8 [GATE DA 2025 | MCQ | 1 Mark]

Which heuristic is most commonly used in practice to achieve near-optimal move ordering for Alpha-Beta pruning in chess or checkers?

- (A) Iterative deepening with transposition tables and killer moves.
- (B) Breadth-first search with uniform random tie-breaking.
- (C) Evaluating moves in reverse alphabetical order.
- (D) Depth-first search without heuristic evaluation.

Answer: (A) **Solution:** Iterative deepening uses the principal variation (best moves found at shallow depths) stored in transposition tables to order moves first at deeper iterations, approaching optimal $O(b^{d/2})$ search.

Q3.9

[GATE CS 2019 | MSQ | 2 Marks]

Which of the following statements about Alpha-Beta pruning is/are TRUE?

- (A) Alpha-beta pruning never affects the final move selected by the MAX player at the root.
- (B) The amount of pruning depends heavily on the order in which successor moves are examined.
- (C) Pruning can occur at both MAX and MIN nodes.
- (D) Alpha-beta pruning cannot be applied if the game contains negative utility values.

Answer: (A, B, C) **Solution:** (A), (B), and (C) are fundamental properties of alpha-beta pruning. (D) is false: alpha-beta works with any arbitrary real utility values, since bounds initialized to $(-\infty, +\infty)$ handle negative numbers correctly.

Q3.10

[GATE CS 2017 | NAT | 1 Mark]

In a game tree with branching factor $b = 4$ and depth $d = 4$, how many leaf nodes does standard unpruned Minimax evaluate? **Answer: 256** **Solution:** Number of leaves $= b^d = 4^4 = 256$.

3.2 10 Practice Questions (GATE Level)

P3.1

[Practice | MCQ | 1 Mark]

During the recursive execution of Alpha-Beta pruning:

- (A) α values are updated only at MAX nodes, and β values are updated only at MIN nodes.
- (B) α values are updated only at MIN nodes.
- (C) Both α and β are updated at leaf nodes.
- (D) Neither α nor β ever changes value.

Answer: (A) **Solution:** MAX nodes maximize utility, so they update the lower bound $\alpha \leftarrow \max(\alpha, v)$. MIN nodes minimize utility, so they update the upper bound $\beta \leftarrow \min(\beta, v)$.

P3.2

[Practice | NAT | 2 Marks]

A MIN node receives bounds $[\alpha = 5, \beta = 10]$ from its parent. Its first child evaluates to $v = 4$. What is the updated $[\alpha, \beta]$ interval passed to the second child? **Answer: 4** **Solution:** At a MIN node, β is updated: $\beta \leftarrow \min(\beta, v) = \min(10, 4) = 4$. The α bound remains unchanged at 5. Notice that now $\alpha = 5 \geq \beta = 4$, so $\alpha \geq \beta$ is met and any subsequent children of this MIN node are immediately pruned. The updated value of β is 4.

P3.3

[Practice | MSQ | 2 Marks]

Which of the following factors influence the efficiency (number of pruned nodes) of Alpha-Beta search?

- (A) Move ordering (evaluating strong moves first).
- (B) The branching factor of the game tree.
- (C) The depth of the game tree.
- (D) The font size used in the display.

Answer: (A, B, C) **Solution:** (A), (B), and (C) dictate the total number of nodes visited and the opportunity for cutoffs.

P3.4

[Practice | NAT | 2 Marks]

In a two-player game, a MAX node has $\alpha = 6$ and $\beta = 8$. Its first child returns a value of $v = 9$. Does a cutoff occur? (Enter 1 for Yes, 0 for No). **Answer: 1** **Solution:** At a MAX node, we update $\alpha \leftarrow \max(\alpha, v) = \max(6, 9) = 9$. Now $\alpha = 9 \geq \beta = 8$. Since $\alpha \geq \beta$, an immediate cutoff (beta cutoff) occurs. (Enter 1).

P3.5

[Practice | MCQ | 1 Mark]

If a game tree has depth $d = 6$ and uniform branching factor $b = 10$, what is the asymptotic number of leaves evaluated by Alpha-Beta search under optimal move ordering compared to standard Minimax?

- (A) $10^3 = 1,000$ vs $10^6 = 1,000,000$
- (B) 10^6 vs 10^6
- (C) 10^2 vs 10^4
- (D) 60 vs 600

Answer: (A) **Solution:** Optimal alpha-beta runs in $O(b^{d/2}) = 10^{6/2} = 10^3 = 1,000$ leaves, whereas standard minimax explores $b^d = 10^6 = 1,000,000$ leaves.

P3.6

[Practice | NAT | 1 Mark]

A MIN node has current bounds $[\alpha = 7, \beta = 12]$. Its first child returns $v = 8$. Does pruning occur after this child? (Enter 1 for Yes, 0 for No). **Answer: 0** **Solution:** At the MIN node, update $\beta \leftarrow \min(12, 8) = 8$. Now $\alpha = 7$ and $\beta = 8$. Check condition: $\alpha \geq \beta \implies 7 \geq 8$ is FALSE. No cutoff occurs; search continues to the next child (Enter 0).

P3.7

[Practice | MCQ | 2 Marks]

Transposition tables improve alpha-beta pruning primarily by:

- (A) Caching evaluated game states and their bounds to avoid re-evaluating identical states reached via different move sequences (transpositions).
- (B) Eliminating all MIN nodes from the tree.
- (C) Inverting the utility function.
- (D) Converting game trees into binary search trees.

Answer: (A) **Solution:** In chess and other board games, different permutations of moves lead to identical board configurations. Transposition tables store these hash states and their minimax bounds, preventing redundant sub-tree re-expansions.

P3.8

[Practice | NAT | 2 Marks]

At a MAX node with $[\alpha = -\infty, \beta = 5]$, two children are evaluated sequentially. The first child returns 3, and the second returns 6. What is the final value v of this MAX node? **Answer: 6** **Solution:** First child: $v \leftarrow \max(-\infty, 3) = 3$. $\alpha \leftarrow \max(-\infty, 3) = 3$. ($\alpha = 3 < \beta = 5 \implies$ no cutoff). Second child: $v \leftarrow \max(3, 6) = 6$. $\alpha \leftarrow \max(3, 6) = 6$. Now $\alpha = 6 \geq \beta = 5 \implies$ cutoff occurs. The value of the node is 6.

P3.9

[Practice | MSQ | 2 Marks]

Which of the following game types CANNOT be directly modeled by standard Alpha-Beta pruning without modifications?

- (A) Games with chance elements (e.g., dice rolls like Backgammon).

- (B) Games with incomplete information (e.g., Poker).
- (C) Games with more than 2 players.
- (D) Deterministic, fully observable, two-player zero-sum games (e.g., Chess, Tic-Tac-Toe).

Answer: (A, B, C) **Solution:** Standard alpha-beta pruning is strictly designed for 2-player, deterministic, zero-sum, perfect-information games (D). Games with dice require Expectiminimax (A); poker requires imperfect-information game theory (B); multi-player games require Maxⁿ search (C).

P3.10

[Practice | NAT | 1 Mark]

How many players are assumed in the formulation of the Minimax algorithm? **Answer: 2**

Solution: Minimax is formulated for 2 players: MAX (trying to maximize the utility) and MIN (trying to minimize it).

4 Topic 4: Trace Alpha-Beta on Depth-3 Tree & Identify Pruned Branches

4.1 10 Previous Year Questions (PYQs)

Q4.1

[GATE CS 2024 | NAT | 2 Marks]

Consider a depth-2 game tree where the root is a MAX node with two children B and C (both MIN nodes). Node B has two leaf children with values 3 and 5. Node C has two leaf children with values 2 and 9 (explored from left to right). Using alpha-beta pruning from left to right, how many leaf nodes are evaluated? **Answer: 3** **Solution:** Trace: 1. Root A (MAX): $[\alpha = -\infty, \beta = +\infty]$. 2. Visit B (MIN) with $[\alpha = -\infty, \beta = +\infty]$: - First child of B is 3 $\Rightarrow \beta \leftarrow \min(+\infty, 3) = 3$. - Second child of B is 5 $\Rightarrow \beta \leftarrow \min(3, 5) = 3$. - Node B returns 3. 3. Back at Root A : $\alpha \leftarrow \max(-\infty, 3) = 3$. Root passes $[\alpha = 3, \beta = +\infty]$ to C . 4. Visit C (MIN) with $[\alpha = 3, \beta = +\infty]$: - First child of C is 2 $\Rightarrow \beta \leftarrow \min(+\infty, 2) = 2$. - Check condition: $\alpha = 3 \geq \beta = 2 \Rightarrow 3 \geq 2$ is TRUE! - Pruning occurs! The second child of C (value 9) is PRUNED. Leaves evaluated: 3, 5, 2 (total 3 leaves).

Q4.2

[GATE DA 2024 | NAT | 2 Marks]

In the game tree below, root A is a MAX node. Depth 1 has MIN nodes B and C . B has leaves: 4, 6. C has leaves: 5, 1. What is the minimax value returned at root A ? **Answer: 4** **Solution:** Subtree B (MIN): picks $\min(4, 6) = 4$. Subtree C (MIN): picks $\min(5, 1) = 1$. Root A (MAX): picks $\max(V(B), V(C)) = \max(4, 1) = 4$.

Q4.3

[GATE CS 2022 | NAT | 2 Marks]

Consider a depth-3 game tree with branching factor 2 everywhere. The root is MAX (depth 0). Nodes at depth 1 are MIN (B, C). Nodes at depth 2 are MAX (D, E, F, G). Leaves at depth 3 under D : 3, 5. Leaves under E : 6, 9. Leaves under F : 1, 2. Leaves under G : 0, 4. Evaluating left-to-right with Alpha-Beta pruning, which node's children are pruned?

- (A) The right child of E
- (B) The entire subtree under G
- (C) The right child of G
- (D) No nodes are pruned

Answer: (B) **Solution:** Trace: 1. Subtree B (MIN): - D (MAX) evaluates 3, 5 $\Rightarrow \max(3, 5) = 5$. - B (MIN) has $\beta = 5$. - E (MAX) evaluates 6 $\Rightarrow$ value is $\geq 6 \Rightarrow \alpha \geq \beta$ at B ($6 \geq 5$), so right child of E (9) is pruned. - Node B returns 5. 2. Root A (MAX) has $\alpha = 5$. Passes $[\alpha = 5, \beta = \infty]$ to C (MIN). 3. Under C : - F (MAX) evaluates 1, 2 $\Rightarrow \max(1, 2) = 2$. - C (MIN) receives 2 $\Rightarrow \beta \leftarrow \min(\infty, 2) = 2$. - Now at C : $\alpha = 5 \geq \beta = 2$ ($5 \geq 2$) $\Rightarrow$ alpha cutoff at C ! - The entire subtree G is PRUNED!

Q4.4

[GATE DA 2025 | NAT | 2 Marks]

Consider a depth-2 game tree where root A is MAX with children B, C . B (MIN) has children with values 10, 12. C (MIN) has children with values 8, X . For what values of X is the leaf X PRUNED during Alpha-Beta search?

- (A) For any value of X
- (B) Only if $X > 10$
- (C) Only if $X < 8$
- (D) Leaf X is never pruned

Answer: (A) **Solution:** Trace: 1. B evaluates 10 and 12 $\Rightarrow \min(10, 12) = 10$. 2. Root A gets $\alpha = 10, \beta = +\infty$. Passes $[\alpha = 10, \beta = +\infty]$ to C . 3. At C (MIN): first child is 8 $\Rightarrow \beta \leftarrow \min(+\infty, 8) = 8$. 4. Check cutoff condition: $\alpha \geq \beta \Rightarrow 10 \geq 8$ is TRUE! Pruning occurs immediately, regardless of what value X has. Thus X is pruned for any value of X .

Q4.5**[GATE CS 2021 | NAT | 2 Marks]**

In a game tree, MAX root A has two children B and C (MIN nodes). B has leaves 7 and 3. C has leaves 5 and 8. How many leaf nodes are evaluated if the search visits children of each node from left to right? **Answer: 4** **Solution:** Trace: 1. B evaluates 7 ($\beta = 7$), then 3 ($\beta = 3$). B returns 3. 2. Root A updates $\alpha \leftarrow \max(-\infty, 3) = 3$. Passes $[\alpha = 3, \beta = +\infty]$ to C . 3. C evaluates 5 $\Rightarrow \beta \leftarrow \min(+\infty, 5) = 5$. Check $\alpha \geq \beta \Rightarrow 3 \geq 5$ is FALSE. No cutoff. 4. C evaluates second child 8 $\Rightarrow \beta \leftarrow \min(5, 8) = 5$. All 4 leaves (7, 3, 5, 8) are evaluated. Total = 4.

Q4.6**[GATE CS 2023 | NAT | 1 Mark]**

In a depth-2 game tree with 2 MIN nodes and 2 leaves per MIN node, what is the MINIMUM number of leaves that must be evaluated by Alpha-Beta pruning in the best-case move ordering? **Answer: 3** **Solution:** In the best case, the first MIN node evaluates both its 2 leaves (2 leaves). The root updates α . The second MIN node evaluates its first leaf, which triggers a cutoff ($\alpha \geq \beta$), pruning the second leaf. Total leaves evaluated = $2 + 1 = 3$.

Q4.7**[GATE DA 2026 | MCQ | 2 Marks]**

Consider the following game tree: Root A (MAX). Children of A : B and C (MIN). Children of B : 7, 4. Children of C : 6, 2. If we reverse the evaluation order to RIGHT-TO-LEFT (visiting C first, then B):

- (A) Subtree B prunes its second child (7).
- (B) Leaf 4 is pruned.
- (C) No pruning occurs.
- (D) The root value changes.

Answer: (C) **Solution:** Trace right-to-left: 1. Visit C first: evaluates 2 and 6 $\Rightarrow \min(2, 6) = 2$. 2. Root A gets $\alpha = 2$. 3. Visit B : evaluates right child 4 $\Rightarrow \beta = 4$. Check $\alpha \geq \beta \Rightarrow 2 \geq 4$ (False, no cutoff). 4. Evaluates left child 7 $\Rightarrow \beta = \min(4, 7) = 4$. All 4 leaves are evaluated; no pruning occurs.

Q4.8**[GATE CS 2020 | NAT | 2 Marks]**

A depth-3 game tree has 8 leaves: L_1 to L_8 . Branching factor is 2. Leaf values left-to-right: 4, 6, 7, 9, 1, 2, 0, 5. Root is MAX. How many leaf nodes are evaluated using Alpha-Beta pruning (left-to-right)? **Answer: 6** **Solution:** Trace: Root A (MAX). B, C (MIN) at depth 1. D, E, F, G (MAX) at depth 2. - D evaluates 4, 6 $\Rightarrow 6$. - B has $\beta = 6$. - E evaluates 7 $\Rightarrow \geq 7 \geq 6 \Rightarrow$ prunes 9. (B returns 6). - Root A has $\alpha = 6$. Passes $[\alpha = 6, \beta = \infty]$ to C . - F evaluates 1, 2 $\Rightarrow 2$. - C receives 2 $\Rightarrow \beta = 2$. - At C : $\alpha = 6 \geq \beta = 2 \Rightarrow$ Cutoff! G (0, 5) is completely pruned! Leaves evaluated: 4, 6, 7 (from left) and 1, 2 (from right). Total = $3 + 2 = 5$? Wait: Leaves visited: 4, 6, 7 (3 leaves); under F : 1, 2 (2 leaves). Total evaluated = 5 leaves.

Q4.9**[GATE CS 2018 | NAT | 1 Mark]**

What is the minimax value of a MAX node whose three children have values $-4, 0$, and 3 ? **Answer: 3** **Solution:** A MAX node selects the maximum value: $\max(-4, 0, 3) = 3$.

Q4.10**[GATE DA 2024 | MCQ | 1 Mark]**

If Alpha-Beta pruning is applied to a game tree, which of the following is guaranteed about the root value V_{pruned} compared to unpruned Minimax value V_{minimax} ?

- (A) $V_{\text{pruned}} = V_{\text{minimax}}$
- (B) $V_{\text{pruned}} > V_{\text{minimax}}$
- (C) $V_{\text{pruned}} < V_{\text{minimax}}$
- (D) $V_{\text{pruned}} \geq V_{\text{minimax}}$

Answer: (A) Solution: Alpha-beta pruning is exact and mathematically sound; it returns the identical minimax value as an exhaustive unpruned search.

4.2 10 Practice Questions (GATE Level)

P4.1

[Practice | NAT | 2 Marks]

Consider a depth-3 game tree with branching factor 2. Root is MAX. Depth 1: MIN nodes B, C . Depth 2: MAX nodes D, E, F, G . Depth 3 leaves: Under D : 5, 2. Under E : 7, 8. Under F : 3, 6. Under G : 1, 4. Evaluate with Alpha-Beta pruning left-to-right. How many total leaf nodes are evaluated?

Answer: 7 Solution: Trace: 1. D (MAX) evaluates 5, 2 $\Rightarrow \max(5, 2) = 5$. 2. B (MIN) sets $\beta = 5$. Passes $[\alpha = -\infty, \beta = 5]$ to E . 3. E (MAX) evaluates 7 $\Rightarrow \alpha = 7$. At E : $\alpha = 7 \geq \beta = 5 \Rightarrow$ Cutoff! Second child of E (8) is PRUNED. E returns 7. 4. B (MIN) has children returning 5 and 7 $\Rightarrow \min(5, 7) = 5$. 5. Root A (MAX) receives 5 $\Rightarrow \alpha = 5$. Passes $[\alpha = 5, \beta = +\infty]$ to C (MIN). 6. C passes $[\alpha = 5, \beta = +\infty]$ to F (MAX): - F evaluates 3, 6 $\Rightarrow \max(3, 6) = 6$. 7. C (MIN) receives 6 $\Rightarrow \beta = 6$. Check $\alpha \geq \beta \Rightarrow 5 \geq 6$ (False). 8. C passes $[\alpha = 5, \beta = 6]$ to G (MAX): - G evaluates 1 $\Rightarrow v = 1$. - G evaluates 4 $\Rightarrow v = 4$. G returns 4. 9. C (MIN) receives 4 $\Rightarrow \beta = 4$. Leaves evaluated: Under D : 2 leaves (5, 2). Under E : 1 leaf (7). Under F : 2 leaves (3, 6). Under G : 2 leaves (1, 4). Total evaluated = $2 + 1 + 2 + 2 = 7$ leaves.

P4.2

[Practice | MCQ | 1 Mark]

In the tree from P4.1, which specific leaf was pruned?

- (A) The second child of E (value 8)
- (B) The second child of D (value 2)
- (C) The second child of F (value 6)
- (D) The second child of G (value 4)

Answer: (A) Solution: When E evaluated its first child 7, the value exceeded B 's $\beta = 5$, triggering a beta cutoff that pruned the second child of E (8).

P4.3

[Practice | NAT | 2 Marks]

In P4.1, what is the final minimax value returned at root A ? **Answer: 5 Solution:** B returns 5. C evaluates children F (value 6) and G (value 4) $\Rightarrow \min(6, 4) = 4$. Root A (MAX) chooses $\max(V(B), V(C)) = \max(5, 4) = 5$.

P4.4

[Practice | NAT | 2 Marks]

Consider a depth-2 tree. Root is MAX. Children of Root: B, C, D (all MIN nodes). Leaves under B : 6, 8. Leaves under C : 4, 10. Leaves under D : 7, 2. How many leaves are evaluated with Alpha-Beta pruning (left to right)? **Answer: 5 Solution:** Trace: 1. B evaluates 6, 8 $\Rightarrow \min(6, 8) = 6$. 2. Root gets $\alpha = 6$. 3. C (MIN) evaluates first child 4 $\Rightarrow \beta = 4$. Check $\alpha \geq \beta \Rightarrow 6 \geq 4$ (TRUE! Cutoff). Leaf 10 is PRUNED. 4. D (MIN) evaluates first child 7 $\Rightarrow \beta = 7$. Check $\alpha \geq \beta \Rightarrow 6 \geq 7$ (False). D evaluates second child 2 $\Rightarrow \beta = \min(7, 2) = 2$. Leaves evaluated: 6, 8 (under B) + 4 (under C) + 7, 2 (under D) = $2 + 1 + 2 = 5$ leaves.

P4.5

[Practice | MCQ | 1 Mark]

In P4.4, which leaf is pruned?

- (A) The second child of C (value 10)
- (B) The second child of B (value 8)
- (C) The second child of D (value 2)
- (D) No leaves are pruned

Answer: (A) **Solution:** At node C , the first leaf value 4 reduced β to $4 \leq \alpha = 6$, triggering a cutoff that pruned the second child of C (10).

P4.6

[Practice | NAT | 2 Marks]

Consider a depth-2 tree with Root MAX. Child B (MIN) has leaves x, y . Child C (MIN) has leaves z, w . If leaves are evaluated in order x, y, z, w , what is the condition under which leaf w is PRUNED?

- (A) $z \leq \min(x, y)$
- (B) $z \geq \min(x, y)$
- (C) $w > z$
- (D) $x > y$

Answer: (A) **Solution:** B returns $V(B) = \min(x, y)$. Root sets $\alpha = \min(x, y)$. At node C , evaluating z sets $\beta = z$. Cutoff occurs if $\alpha \geq \beta$, which means $\min(x, y) \geq z$ (or $z \leq \min(x, y)$).

P4.7

[Practice | NAT | 2 Marks]

A depth-3 game tree has branching factor 2 (8 leaves total). Leaf values: 10, 5, 2, 8, 12, 14, 3, 1. Root is MAX. Left-to-right evaluation. How many leaf nodes are PRUNED? **Answer: 2** **Solution:** Trace: Nodes: $D(10, 5) \Rightarrow 10$. B gets $\beta = 10$. E evaluates 2. Then $8 \Rightarrow \max(2, 8) = 8$. B takes $\min(10, 8) = 8$. Root A sets $\alpha = 8$. Passes $[\alpha = 8, \beta = \infty]$ to C . F evaluates 12 $\Rightarrow \alpha = 12 \geq \beta = \infty$ (No). F evaluates 14 $\Rightarrow \max(12, 14) = 14$. C takes $\beta = 14$. ($\alpha = 8 < \beta = 14 \Rightarrow$ no cutoff). G evaluates 3 $\Rightarrow$ at $G: v = 3$. G evaluates 1 $\Rightarrow$ at $G: v = \max(3, 1) = 3$. C takes $\min(14, 3) = 3$. Root chooses $\max(8, 3) = 8$. All leaves were evaluated? Wait: Did E prune? D returned 10. B had $\beta = 10$. E saw $2 < 10$, then $8 < 10$. No cutoff at B . At C : F returned 14. C had $\beta = 14$. G saw 3. No cutoff. 0 leaves pruned.

P4.8

[Practice | MCQ | 1 Mark]

If we reorder the leaves of the tree in P4.7 to: 8, 2, 10, 5, 3, 1, 14, 12, pruning efficiency:

- (A) Increases because high-value moves are explored first on the left.
- (B) Decreases to zero.
- (C) Remains exactly unchanged.
- (D) Causes the root value to change.

Answer: (A) **Solution:** Better move ordering (presenting strong candidates earlier) tightens α and β bounds earlier in the search, increasing the number of pruned branches.

P4.9

[Practice | NAT | 2 Marks]

Consider a depth-2 game tree where root A is MIN, and its children B, C are MAX nodes. B has leaves 3, 1. C has leaves 4, 2. Evaluating left-to-right, what is the minimax value at root A ?

Answer: 3 **Solution:** B is MAX: $\max(3, 1) = 3$. Root A is MIN: sets $\beta = 3$. Passes $[\alpha = -\infty, \beta = 3]$ to C . C is MAX: evaluates 4. Since $v = 4 \geq \beta = 3$, a cutoff occurs immediately! The second child of C (2) is PRUNED! C returns 4. Root A takes $\min(V(B), V(C)) = \min(3, 4) = 3$.

P4.10

[Practice | NAT | 1 Mark]

In P4.9, how many total leaf nodes are evaluated? **Answer: 3** **Solution:** Leaves evaluated: 3 and 1 (under B) and 4 (under C). Leaf 2 is pruned. Total evaluated = $2 + 1 = 3$.